

Compliance Sheet

PROJECT REFERENCE	REV.	LANG.	
The Irrigation and Domestic pump systems	A	en	

2.6 **Variable speed drives (VSDS)**

2.6.1 **General:**

- **This part of the Specification describes the general requirements for the variable speed drives, the VSDs. The nominal values, the standard documents and the drive's minimum performance are defined in this part. The VSD does not include motors in this specification. The Specification uses the term motor unit which means a combination of the VSD and the motor.**

- **If the Project specific part of the Specification is in contradiction with the other parts of the document, the Project specific document shall apply.**

- **Requirements for the Manufacturer:**
 - Country of Origin/Make: The complete VSD as a unit should be manufactured in its origin as approved by Engineer.
 - **Comply, Schneider Electric is one of the approved vendors**
 - Certifications: The frequency converter manufacturer shall have a valid ISO 9001 certification and an applicable quality assurance system. The manufacturer shall have the environment certification ISO 14001.
 - **Comply, ATV680 drives already comply with this ISO certification**
 - Experience: The manufacturer shall have a minimum of 5 years' experience in frequency converter manufacturing and have adequate business volume in order to provide credibility in his commitments and a capability of long-term support. The manufacturer shall prove his experience by quoting references of units in the specified power and voltage range.
 - **Comply as we have a very huge reference list on Egypt**

- **Local support:**
 - The supplier shall have a permanent representative office with a trained and skilled support staff, in order to prove his commitment for local support and to provide a channel for communication. The local representatives shall be easily accessible and shall be able to have a technical team at the site within 24 hours' notice, 7 days of the week.
 - **Comply**
 - The engineers employed by the supplier's regional office shall be certified by the manufacturer and provide start-up service including physical inspection of the drive, connected wiring and final adjustments, to ensure that the VSD meets the required performance.
 - **Comply**

- The supplier shall be able to give basic drives training to the Customer's engineers, preferably on the site but anyway, in the country where the customer's site is. The training shall, as a minimum, include system concepts and basic troubleshooting. The supplier shall also be capable of solving most VSD problems quickly. He shall also have 24-hour support from the drives' factory, to avoid any delays during service or repair work on the site.
 - **Comply**
- The manufacturer shall be able to offer commissioning and certification of the installation of the drive to be done by the local office.
 - **Comply**
- The most common spare parts like fuses, IGBTs as well as main control and IO boards shall be available in 48 hours from the notification through a regional service center of the supplier. The more rarely used spare parts should be available in maximum 7 days on site.
 - **Comply**
- Basic requirements for the VSDs
 - **Comply**

▪ **General requirements:**

- Provide Variable speed drives suitable for all types and sizes of pumps and other equipment as shown on the drawings and schedules.
 - **Comply**
- The VSD shall be of modern design, user friendly and be simple to install, commission and maintain.
 - **Comply as The ATV680 includes an active input bridge and leads a sinusoidal main current. So, it is an alternative to active and passive filters. State-of-the-art components, a new control concept as well as a top-quality filter module reduce the total harmonic distortion THD(i) of the frequency inverter to a value less than 2...5 % and by this way it fulfills the requirements according to the recommendations in IEEE 519 to reduce the current harmonics in the mains. Also, in case of operation with partial load (from 30 % P_n) the ATV680 leads to a sinusoidal main current with a power factor $\cos \Phi \approx 1$.**
- The VSD shall be able to start and control the speed of a standard squirrel cage induction AC motor.
 - **Comply**

- The VSDs shall be CE marked and comply with IEC standards.
 - Comply, the frequency inverters have a CE marking on the rating plate. However, to achieve the corresponding limits it is necessary to observe the installation regulations, superior and regional standards and directives as well as the directives listed below.
 - All devices and drives of electric drive engineering may cause electromagnetic interferences and otherwise they may be influenced by such interferences. Therefore, they are subject to the **EMC directive 2004/108/EC**.
- The materials used in the VSD as a complete unit shall be recyclable, non- toxic and flame retardant.
 - Comply
- The VSD shall be equipped with a standard communication protocol for interface with BMS and SCADA /PLC system.
 - Comply, As ATV680 has a built in Ethernet, Modbus communication
- The VSD shall be a digitally controlled drive, using, at least, the pulse width modulation (PWM) with flux vector control, direct torque control (DTC). It shall have IGBT's in the inverter section throughout the power range.
 - Comply using PWM
- The cable type and its maximum length between the VSD and Motor to be according to manufacturer recommendation and Engineer Approval.
 - Cable length is up to 150 mt from VSD to motor with a recommended shielded cable
- **Operating conditions:**
 - Rated Input Voltage: 380V- 400V, three-phase
 - Comply. As per the technical offer
 - Input AC Voltage Tolerance: ± 10 percent of input voltage rating
 - Comply. As per the technical offer
 - Input voltage unbalance: $< 3.5\%$
 - Comply. As per the technical offer
 - Rated Input Frequency: 50Hz $\pm 5\%$.
 - Comply. As per the technical offer
 - Fundamental Power Factor: 0.97 or better at full load.
 - Comply. Additionally, the $\cos \Phi \approx 1$ in each load situation helps to reduce the load of the mains
 - Efficiency: % 98 percent at full load.
 - Comply with 98 % efficiency

- Output Voltage: 0- UN, 3 phase.
 - **Comply**
- Output Frequency Range: 0 to 500 Hz
 - **Comply**
- Output Frequency Resolution: 0.01 Hz.
 - **Comply**
- Acceleration and deceleration ramps shall be independently adjustable, and the inverter shall be capable of riding a 500 millisecond mains interruption without causing tripping.
 - **Comply**
- **Overload ability (by load type):**
 - Constant Torque: Up to 150 percent of nominal current for 1 min every 5 mins.
 - **As we are working on pump applications, the design of the VSD is on Normal duty selection: 110 % for 60 s per 10 minutes**
 - Variable Torque (pump & fan): No overload ability required.
 - **Comply .110 % for 60 s per 10 minutes**
 - Ambient Temperature: 45 deg. C.
 - **Comply with derating factor applied**
 - Installation Altitude: Sea level up to 1000m
 - **Comply without derating**
 - Max. Relative Humidity: 90 percent, non-condensing.
 - **Comply. no condensation inside the enclosure, max. 95 % relative humidity**
- **The Drive shall provide the following minimum protection. Occurrence of any of these faults / protections shall cause electronic shut down of the variable speed (frequency) drive controller without blowing of any fuses or tripping of circuit breaker:**
 - Motor phase to phase fault.
 - **Comply**
 - Motor phase to earth fault -- continuous during operation.
 - **Comply**
 - Overvoltage – Under voltage - Inverter overheat - Motor overheat - Stall protection - Loss of control signal - Loss of auxiliary control voltage - Current limit.
 - **Comply**

- **Main Protections: Over current, short circuit, input/output phase loss, motor overload and under load, over/under-voltage, over speed, over temperature, motor stall, another internal fault.**
 - **Comply**
- **The VSD shall be able to give a 100 percent output current continuously in the above-specified conditions. In order to ensure that the drive can provide the required output current in the specified ambient conditions, the manufacturer shall inform of the required de rating, if the ambient temperature given in the Project specific Specification is higher than 50 deg.C or if the installation altitude is more than 1000m above the sea level. The de rating factor shall be specified so that neither the lifetime of the VSD nor the unit's performance, overload ability included, nor the reliability of the VSD shall suffer.**
 - **Design is made based on 45 C ambient temp, So the derating is already applied after 40 C**
- **Storage conditions (in the protective package):**
 - Ambient Temperature: -40° to +60°C
 - Corrosion Level of the Cooling Air Chemical Gases: IEC 60721-3-3, class 1C2. Solid Particles: IEC 60721-3-3, class 1S3.
 - Max. Vibration Level (IEC 60068-2-6) 2 to 9 Hz: 1.5 mm. 9 to 200Hz: 5 m/s².
 - Shock (IEC 60068-2-29): Max. 100 m/s, 11ms.
 - Freefall: 250mm for weight under 100kg. 100 mm for weight over 100 kg.
 - Ambient conditions
 - **Ambient temperature -10...+50 °C (below 0 °C with additional enclosure heating, above +40 °C with derating). 3K3 according to IEC/EN 60721-3-3**
 - **Storage / Transport temperature -25...+70 °C**
 - **Protection degree Door closed: Door open: IP23 (optionally enclosure design IP54) IP2x**
 - **Environmental class / Humidity Class 3K3 in accordance with IEC/EN 60721-3-3 / no condensation inside the enclosure, max. 95 % relative humidity**
 - **Altitude Up to 1000 m no derating necessary 1000...2000 m derating of 1 % / 100 m (for all types of mains) 2000...3800 m derating of 1 % / 100 m (only TT/TN, IT) 3800...4800 m derating of 1 % / 100 m (only TT/TN)**
 - **Allowed pollution: Pollution degree IP23: 2 according to EN 61800-5-1
Pollution degree IP54 (optional): 3 according to EN 61800-5-1
Chemical / mechanical classification: 3C3 and 3S3 according to EN 60721-3-3**
 - **Protection class: Class 1 according to EN 61800-5-1**

- **Panel Design Specs:**

- Standards: IEC 61439-1, EN 61439 & VDE 660 Part 500.
- Protection Class: IP 54 .
 - **Comply, IP54**
- Cabinet access: From front.
- Cable entry and exit: Bottom entry as standard. Both bottom and top entry have to be possible.
- Color: Grey RAL 7032 with White RAL 9003 inside the panel/s.
 - **Color: RAL 7022**
- User interface: For user interface refers to control specification

- **Software features:**

- Power loss ride-through: The drive shall have a power loss ride-through capability. This means that the drive controls should stay alive during a power loss by means of the energy stored in the load. The ride through time shall be the longer the higher the kinetic energy of the load is. The motor shall be magnetized as long as there is kinetic energy in the system.
 - **Comply**
- Flux Optimization: The VSD shall have a built-in automatic flux optimization function. The flux optimization function minimizes the sum of the magnetizing current and the load current so that the drive can still follow the given reference. This feature reduces energy consumption and motor noise when driving at less than the nominal load.
 - **Comply**
- Critical speed jump-over: The VSD shall have programmable skip speeds to jump over critical resonance speeds. If the speed reference is in the critical speed area, it is ignored, and the latest speed reference is maintained. 3 programmable critical speeds at least shall be available.
 - **Comply with 3 skip frequencies**
- Current/speed limiting: In case the acceleration or deceleration ramps are too fast for the drive capacity, the drive shall be able to automatically reduce the ramp to prevent tripping. Also, in case of transient overloads the drive shall automatically reduce speed to prevent an over current trip, if the drive capacity is not sufficient to handle the load.
 - **Comply**

- PID-controller the drive shall have a built-in PID-controller for control of the customer process.
 - **Comply**
- Restart in the event of a fault trip due to over voltage, over current or loss of analog signal, the VSD shall be programmable to attempt an automatic restart. For safety reasons, the maximum number of attempts shall be five (selectable) within a short time. If the fault does not clear after the attempts, the drive shall lock out.
 - **Comply**

▪ **Environmental effects:**

- Harmonic Distortion: The VSD to be equipped with built-in AC or DC chokes to minimize the total harmonic distortion (THD). The THD limit for current and voltage to be according to IEEE 519 standard limit. However, the VSD manufacturer shall submit to the Contractor the VSD harmonic spectrum for the Project specific supply network. The spectrum shall be used in the design of appropriate harmonic filters. The single harmonics shall be presented up to 25th harmonic and the THD has to be calculated taking into consideration harmonics up to 40th harmonic.
 - **Comply using the ATV680 LH drives with two active and passive filters.**
 - **State-of-the-art components, a new control concept as well as a top-quality filter module reduce the total harmonic distortion THD(i) of the frequency inverter to a value less than 2...5 % and by this way it fulfills the requirements according to the recommendations in IEEE 519 to reduce the current harmonics in the mains.**
 - **Also, in the case of operation with partial load (from 30 % P_n) the ATV680 leads a sinusoidal main current with a power factor $\cos \Phi \approx 1$.**
- EMC Regulations and Compatibility/C-Tick: The supplied VSDs shall carry the CE mark indicating that they comply with the essential requirements of the relevant EU directives. The VSDs shall meet the requirements set in IEC EN 61800-3 for Industrial Low-Voltage Networks. If the project- specific specification states that the requirements for Public Low-Voltage Networks stated in IEC EN 61800-3 must be met, the supplier shall be able to provide such units in earthed networks. If separate EMC filters are required, they shall be of built-in type.
 - **Comply**
 - **All devices and drives of electric drive engineering may cause electromagnetic interferences and otherwise they may be influenced by such interferences. Therefore, they are subject to the EMC directive 2004/108/EC.**
 - **The frequency inverters have an operating voltage which is clearly in the range of 50...1000 V AC or 75...1500 V DC. Therefore, they are also subject to the Low Voltage Directive 2006/95/EC.**
 - **Because of the EMC filters which are built into the frequency inverters they are in conformity with EN 61800-3 and EN 61800-5-1.**

- A detailed description and other directions to maintain the EMC Compatibility during the installation of the VSD and associated field cables and connections, shall be given by the Supplier in conformance with the EMC Directives or C-Tick. The Contractor shall follow the directions during installation, in order to achieve attenuation of the RFI.
- **Comply**
- **EMC immunity According to EN 61800-3, second environment (EN 61000-4-2; EN 61000-4-3; EN 61000-4-4; EN 61000-4-5; EN 61000-4-6)**
- **EMC emission in accordance with product standard EN 61800-3, second environment, category C3**
- Audible Noise: The full load audible noise of the frequency converter as measured at a distance of 1000mm, shall not exceed 70 dB(A) in 200 kW applications and below. Above 200 kW, the full load audible noise shall not exceed 78 dB(A). If the frequency converter is installed in a cabinet and requires a separate cooling fan, these limits also include the noise of the additional cooling fan. This requirement is made to keep the electrical room quiet so that it is not necessary to use hearing protection. The audible noise of the motor should also be minimized. For that purpose, the switching frequency of the frequency converter shall be at least 2 kHz throughout the power range.
- **Comply**
- Efficiency: The full load efficiency of the VSD shall be at least 98 percent including all the additional equipment, which is needed to meet the Specification.
- **Comply**